

SENTINEL-XAI: Hybrid Explainable AI for Spacecraft Health Monitoring and Fault Diagnosis

Independent Technical Research Project

Sabrina Benghenane

Simulation-based research prototype | 120 independent Monte Carlo scenarios | Frozen final system

Abstract

Autonomous spacecraft health monitoring must detect abnormal behavior, identify the likely fault, quantify uncertainty, and provide evidence engineers can inspect. This project presents SENTINEL-XAI, a hybrid framework combining engineering rules, physics/model-based residual monitoring, unsupervised Isolation Forest anomaly evidence, supervised Random Forest diagnosis, confidence-aware decision logic, and SHAP explanations. Six fault modes were modeled across power, thermal, reaction-wheel, and sensor subsystems. In development, the hybrid detector achieved precision 1.0000, recall 0.7487, and F1 0.8563, exceeding the rule-based (0.7207), model-based (0.7870), and unsupervised-ML (0.2502) F1 scores. The diagnosis model achieved 93.32% accuracy and 0.9419 macro F1. The complete frozen system was then evaluated on 120 independently randomized fault scenarios without retraining or confidence-threshold retuning. It retained hybrid F1 0.8294 and diagnosis macro F1 0.9003. Every scenario produced at least one correct accepted diagnosis. SHAP faithfulness also transferred: ablating the top five SHAP features reduced predicted-class probability by 0.6305 on average versus 0.0148 for random features. The study supports heterogeneous evidence fusion for robust and interpretable spacecraft fault monitoring within the tested simulation envelope, while exposing confidence calibration and battery-degradation diagnosis as remaining limitations.

Keywords: spacecraft health monitoring, FDIR, hybrid AI, fault diagnosis, Isolation Forest, Random Forest, SHAP, Monte Carlo validation.

1. Problem and Project Objective

Spacecraft operate with limited maintenance opportunities and may need to react autonomously when telemetry departs from expected behavior. A useful onboard monitoring system should therefore answer three questions: *Is something wrong? What is the likely fault? Why does the system believe that?* Traditional rule-based logic is transparent but can miss gradual faults; physical residual monitoring is interpretable but depends on model coverage; machine learning can capture multivariate patterns but may generalize poorly or behave as a black box [1, 2, 3, 9].

The objective of SENTINEL-XAI was to test whether these approaches can be combined so that their strengths compensate for one another. The project was developed incrementally: nominal simulation, controlled fault injection, rule baseline, model-based monitoring, unsupervised anomaly detection, hybrid fusion, supervised diagnosis, confidence gating, explainable AI, and finally independent Monte Carlo robustness testing.

The main research contribution is the end-to-end architecture rather than a single classifier: telemetry is converted into rule, physics, and ML evidence; the evidence is fused for detection; a supervised model identifies the fault class; a frozen confidence gate separates accepted from uncertain diagnoses; and SHAP exposes the features supporting each decision.

2. Simulation and Fault Modeling

The simulator represents four interacting spacecraft health-monitoring domains: electrical power, battery state, thermal behavior, and reaction-wheel dynamics, with noisy measured telemetry generated from underlying simulated states. The nominal six-hour dataset contains 361 one-minute samples, and an additional 24-hour nominal run was used for long-duration validation and anomaly-model calibration.

The six injected fault modes were: solar-array degradation, battery degradation, thermal anomaly,

reaction-wheel degradation, battery-voltage sensor drift, and telemetry dropout. Faults support abrupt and gradual profiles. The independent robustness campaign later randomized onset between 1.5 and 3.5 hours, severity from 0.20 to 0.80, gradual ramp duration from 0.20 to 1.00 hour, and sensor-noise scale from 0.75 to 1.50.

Table 1: Faults and principal simulated signatures.

Fault	Primary signature
Solar-array degradation	Reduced solar generation and reduced charging margin.
Battery degradation	Degraded battery behavior and weaker energy retention.
Thermal anomaly	Increased battery/electronics temperatures relative to nominal reference.
Reaction-wheel degradation	Increased wheel-current and wheel-temperature evidence.
Battery-voltage sensor drift	Bias in measured battery voltage without the same physical change in true voltage.
Telemetry dropout	Missing measured telemetry during the active fault interval.

3. SENTINEL-XAI Architecture

Figure 1 shows the final frozen system. The rule branch applies engineering thresholds and missing-data logic. The model branch uses a battery EKF plus thermal and reaction-wheel residual monitoring. The ML branch uses four subsystem-specific Isolation Forest models trained on nominal data only. Hybrid fusion combines these three evidence sources. A Random Forest then performs seven-class diagnosis (nominal plus six faults), followed by a fixed confidence threshold of 0.70 and SHAP explanation.

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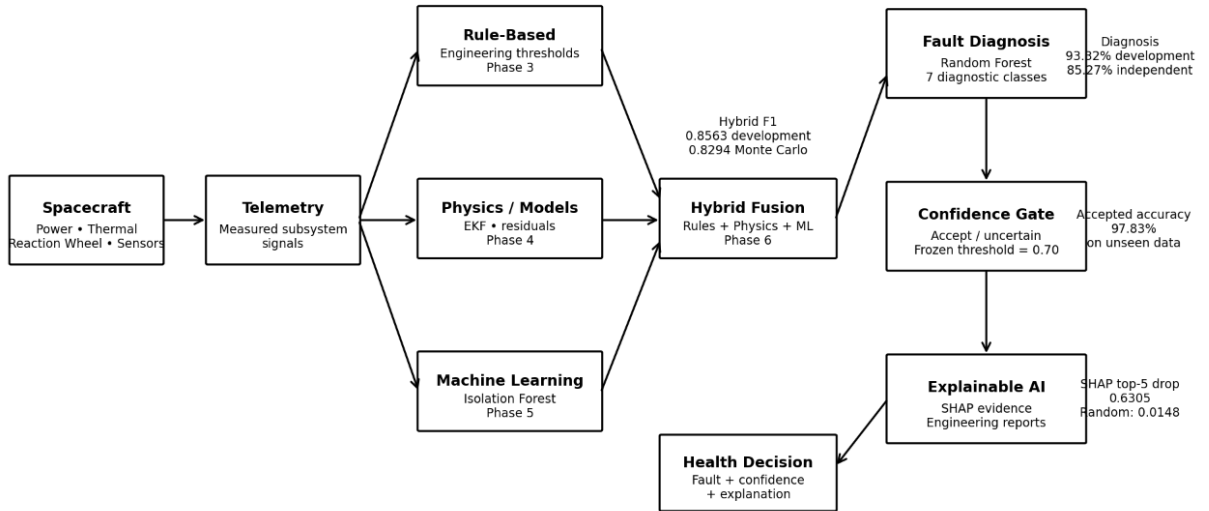


Figure 1: Final SENTINEL-XAI architecture and headline validated results.

The design deliberately avoids using hidden simulation truth variables as operational diagnosis features. The final diagnostic vector contains 57 measured or derived features, including telemetry, residuals, detector predictions, confidence/evidence signals, and missing-data indicators.

4. Detection and Diagnosis Methods

Rule-based baseline. Engineering conditions monitor low solar production in sunlight, temperature limits, wheel-current behavior, state of charge, voltage residuals, and telemetry dropout. The method produced zero development false alarms but missed many weak/gradual faults.

Model-based monitoring. A battery EKF estimates state of charge and voltage innovation, while thermal and reaction-wheel monitors compare observations against nominal reference behavior.

Residuals take the generic form

$$r_k = y_k - \hat{y}_k, \quad (1)$$

with rolling-window and persistence logic suppressing isolated excursions.

Unsupervised ML. Four subsystem Isolation Forests [5] use 20 measured/derived features. The final configuration uses 300 trees, nominal-only training, regime-balanced calibration, a 99th-percentile threshold, and three-sample persistence. This branch was intentionally retained despite weak standalone recall because it provides complementary evidence, especially for unusual telemetry patterns and dropout.

Hybrid fusion. The fusion layer combines rule alarms, model-based alarms, and persistent ML evidence. It improved overall detection F1 while retaining very high precision.

Supervised diagnosis. A Random Forest [6] maps the 57-feature diagnostic representation to seven health classes. Development evaluation used a blocked chronological split within class to reduce optimistic leakage from adjacent time samples.

Confidence and XAI. The maximum class probability is compared with a frozen threshold of 0.70. Accepted predictions are explained with SHAP [7]. Explanation faithfulness is measured by removing the five most important SHAP-ranked features and comparing the probability drop with random five-feature ablation. Stability is measured using top-feature Jaccard overlap and class-consensus overlap.

5. Development Results

The progression from standalone methods to hybrid fusion is summarized in Table 2 and Figure 2. The rule baseline achieved F1 0.7207; the model-based detector improved to 0.7870; the unsupervised detector alone reached only 0.2502; and the hybrid system achieved F1 0.8563 with precision 1.0000 and recall 0.7487. Mean detection/diagnosis delay also improved from 33.7 minutes for the rule baseline to 15.5 minutes for the hybrid detector.

Table 2: Development detection comparison.

Method	Precision	Recall	F1	Delay (min)
Rule-based	1.0000	0.5634	0.7207	33.7
Model-based	1.0000	0.6488	0.7870	16.7
Isolation Forest	0.9174	0.1448	0.2502	50.0
Hybrid fusion	1.0000	0.7487	0.8563	15.5

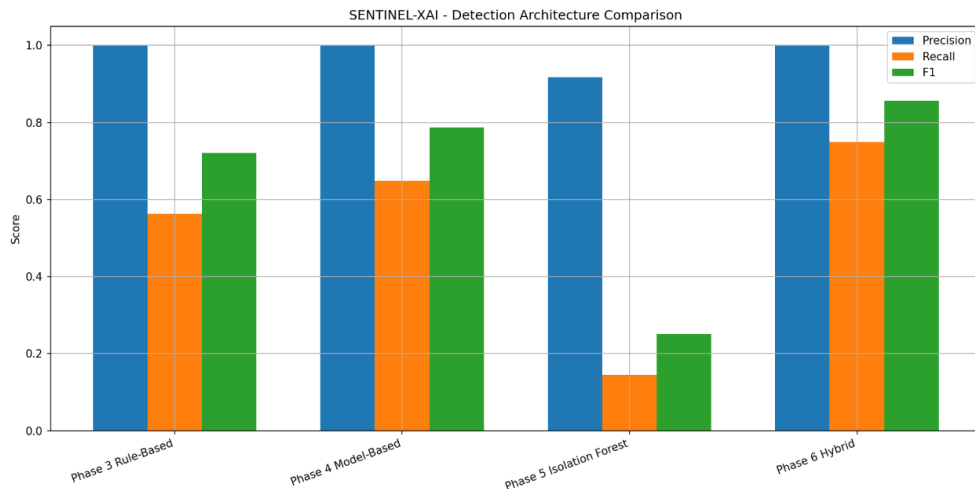


Figure 2: Development comparison of rule-based, model-based, unsupervised ML, and hybrid detection.

The supervised diagnosis layer achieved 0.9332 raw accuracy, 0.9447 balanced accuracy, and 0.9419 macro F1 on the blocked development evaluation. The development confidence gate accepted 68.70%

of predictions and achieved 1.0000 accepted accuracy in that development setting. These figures were treated as development results only; the threshold was frozen before independent testing.

6. Independent Monte Carlo Robustness

The independent evaluation used 120 new scenarios: 20 for each fault class. The system was not retrained, the 0.70 confidence threshold was not changed, and SHAP settings were not tuned after observing the test results. Randomization covered fault onset, severity, profile, gradual ramp duration, sensor seed, and measurement-noise scale.

On this broader distribution, the hybrid detector retained precision 0.9970, recall 0.7100, and F1 0.8294. The diagnosis model achieved 0.8527 raw accuracy, 0.8541 balanced accuracy, and 0.9003 macro F1. Every one of the 120 scenarios produced at least one correct accepted diagnosis. Mean correct-diagnosis delay was 19.45 minutes and the median was 12.55 minutes.

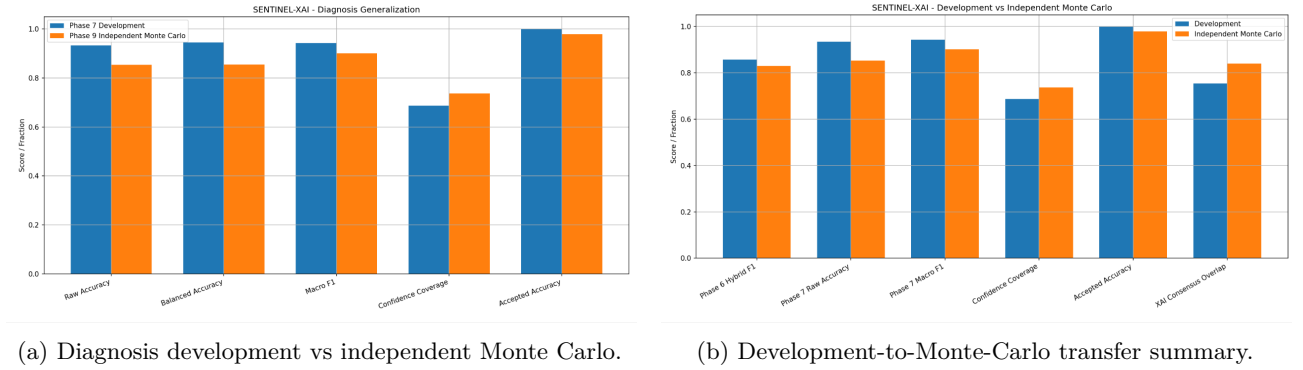


Figure 3: Generalization of diagnosis and system-level metrics.

Per-fault analysis identified battery degradation as the main weakness. Its raw diagnosis accuracy was 0.7507, confidence coverage only 0.3373, hybrid recall 0.3039, end-to-end correct fraction 0.1558, and mean correct-diagnosis delay 51.99 minutes. By contrast, telemetry dropout remained nearly perfect, and thermal anomalies retained strong performance. Gradual faults were more difficult than abrupt faults: hybrid recall was approximately 0.695 for gradual scenarios versus 0.744 for abrupt scenarios. Sensor-noise scaling from 0.75 to 1.50 had a comparatively modest effect within the tested range.

Table 3: Independent per-fault robustness summary.

Fault	Raw Acc.	Conf. Cov.	Hybrid Recall	Delay (min)
Solar degradation	0.9070	0.6978	0.5484	10.33
Battery degradation	0.7507	0.3373	0.3039	51.99
Thermal anomaly	0.9183	0.8957	0.8824	24.04
Reaction-wheel degradation	0.7649	0.7036	0.7195	17.15
Voltage sensor drift	0.7840	0.8041	0.8061	12.73
Telemetry dropout	1.0000	1.0000	1.0000	0.48

7. Confidence Transfer and Explainability

Independent confidence testing used 25,221 active-fault samples. At the frozen 0.70 threshold, coverage was 0.7371, accepted accuracy 0.9783, and error-capture rate 0.8913. The original development targets were accepted accuracy at least 0.99, error capture at least 0.95, and coverage at least 0.60. Thus, coverage transferred successfully, while accepted accuracy and error capture did not. The threshold was intentionally left unchanged; descriptive tests at 0.80 and 0.90 were reported only to characterize the trade-off, not to retune the test set.

This limitation is important because confidence alone was not foolproof: the maximum confidence among independent classifier errors reached 0.9897. Therefore, a high classifier probability should not be interpreted as proof of correctness.

The XAI results were more stable. On accepted, correct development predictions, removing the top

five SHAP features reduced predicted-class probability by 0.7026 on average versus 0.0145 for random features. In the independent Monte Carlo campaign, the corresponding values were 0.6305 and 0.0148, and SHAP top-five ablation beat random ablation in all 120 tested scenarios. Mean within-class pairwise top-five Jaccard overlap was 0.6876, Monte Carlo consensus overlap was 0.8400, and overlap between development and Monte Carlo class top-five explanations was 0.7333.

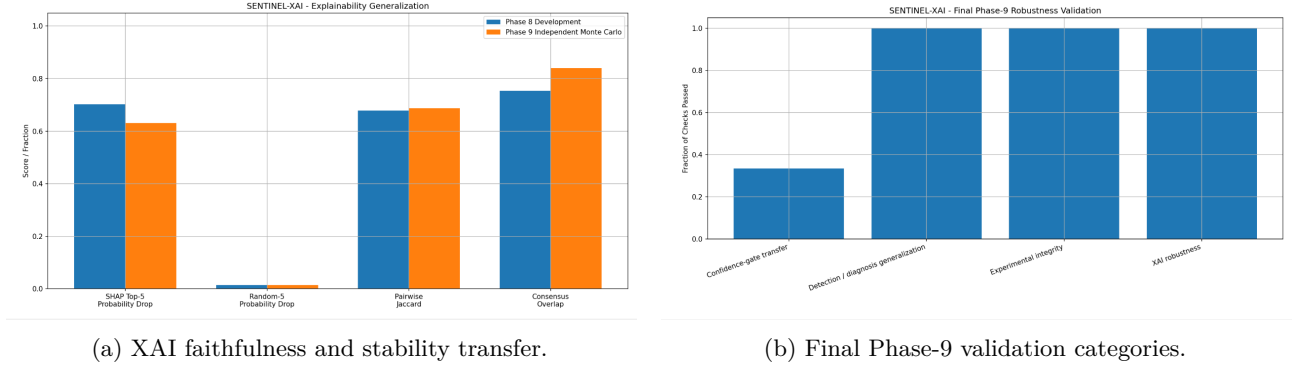


Figure 4: Independent explainability and final robustness validation.

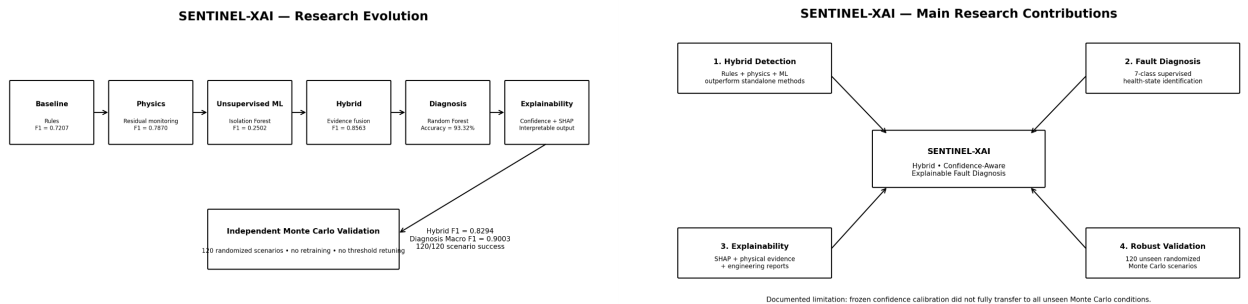
SHAP is used here to explain the Random Forest decision, not to prove physical causality. The engineering interpretation remains grounded in measured telemetry, residuals, detector outputs, and subsystem knowledge.

8. Discussion, Limitations, and Future Work

The strongest result is not that one ML algorithm solved spacecraft diagnosis. In fact, the standalone unsupervised detector performed poorly for several gradual physical faults. The project instead shows the value of combining evidence with different failure modes: rules contributed conservative high-precision logic, model-based residuals provided physically meaningful deviation signals, ML added distribution-sensitive anomaly evidence, and the supervised layer integrated these signals into explicit fault labels.

The independent study also prevented an overly optimistic conclusion. Development diagnosis accuracy fell from 0.9332 to 0.8527 under randomized conditions, and the confidence gate did not fully transfer. Battery degradation remained difficult because its signature overlaps nominal power-system evolution and often develops gradually. The project therefore demonstrates useful robustness within a controlled simulation envelope, not universal robustness or flight readiness.

Key limitations are: simulated rather than flight telemetry; single-fault scenarios rather than simultaneous faults; simplified physical models; no hardware-in-the-loop validation; a confidence threshold selected on development data; and a limited operating envelope. Future work should add an independent calibration set, probability-calibration methods, more operating regimes, simultaneous/compound faults, hardware-in-the-loop tests, and real spacecraft telemetry where available.



(a) Evolution from baselines to independent validation.

(b) Final research contribution map.

Figure 5: Project evolution and consolidated contributions.

9. Conclusion

SENTINEL-XAI developed and validated a hybrid, confidence-aware, and explainable spacecraft health-monitoring prototype. Relative to the individual development baselines, hybrid fusion produced the strongest detection F1 (0.8563). A supervised diagnostic layer achieved 0.9419 development macro F1 and retained 0.9003 macro F1 on 120 independent randomized scenarios. Explanation faithfulness remained strong under the same independent evaluation, while confidence transfer exposed a genuine limitation that was preserved rather than tuned away.

The resulting system provides a coherent research answer to the original problem: spacecraft telemetry can be transformed into a fault decision that combines rules, physics/model residuals, anomaly evidence, learned diagnosis, uncertainty handling, and interpretable feature evidence. The results justify further investigation of the architecture under higher-fidelity and real operational conditions, but they should not be interpreted as spacecraft certification or flight qualification.

Project Reproducibility Status

The completed repository contains 56 experiment scripts, 131 result tables, 109 figures, four saved models, 12 research reports, and 120 Monte Carlo telemetry files. The final repository audit found all 53 required project artifacts and passed all 9 final validation checks. Frozen model files, a README, requirements file, and reproducibility guide are included in the project repository.

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